# Selection

CSE 101 Section 03
Introduction to Algorithms and Computing
Fall 2022
© Royal Military College of Canada

# Selection

If you are in the kitchen:

get a snack

If you are taking CSE 101 and MAE 101

    stand on your head

else

    do a cartwheel

```
If you are wearing green or orange
        sing a song
else if you are wearing blue
        chirp like a bird
else if you are wearing pants
        bark like a dog
else
        quack like a duck
```

# The if statement

- Matlab syntax:

```
if condition
     statement
     statement
end
```

- First line known as the `if` clause
  - Includes the keyword `if` followed by condition
    - The condition can be `true` or `false`
    - When the `if` statement executes, the condition is tested, and if it is true the block of statements is executed. otherwise, the block of statements is skipped

# If condition ?

- The condition in the if statement is a *Boolean expression*, i.e. expression tested to determine if it is true or false
  - Example: a > b
    - `true` if a is greater than b; `false` otherwise

- Conditions rely on *relational operators* (covered in previous lecture) to determines whether a specific relationship exists between two values.

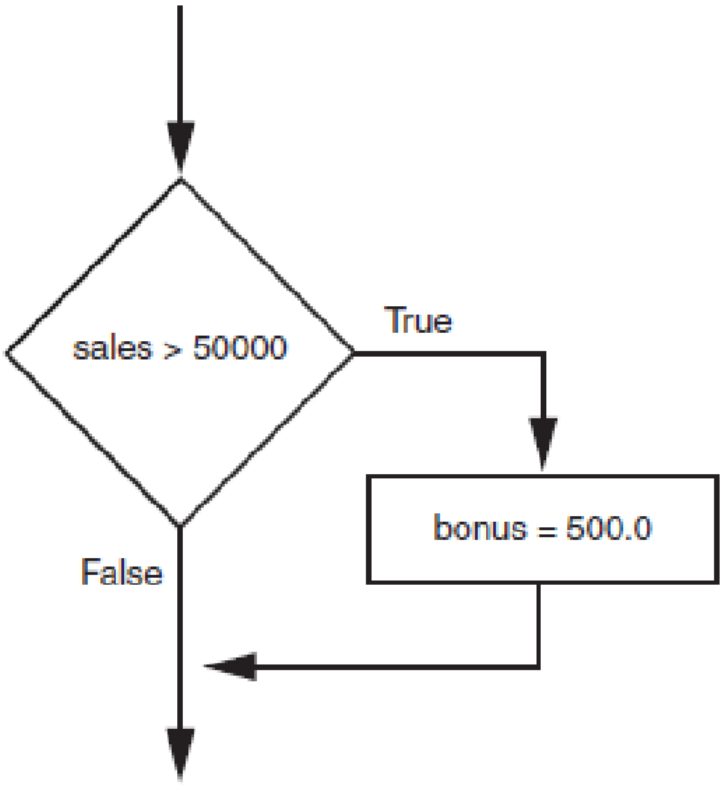
# RECALL: Operations: Relational

- Relational operators compare the elements in two arrays and return logical `true` or `false` values to indicate where the relation holds.

| | | |
|---|---|---|
| == | Determine equality | 3 == 2 |
| >= | Determine greater than or equal to | 2 >= 1.5 |
| > | Determine greater than | 2 > 2 |
| <= | Determine less than or equal to | 5 <= 7 |
| < | Determine less than | 3 < 3 |
| ~= | Determine inequality | 3 ~= 3 |

# Example

```
sales = 100000;
if sales > 50000
    bonus = 500
end
```



# Example of testing equality

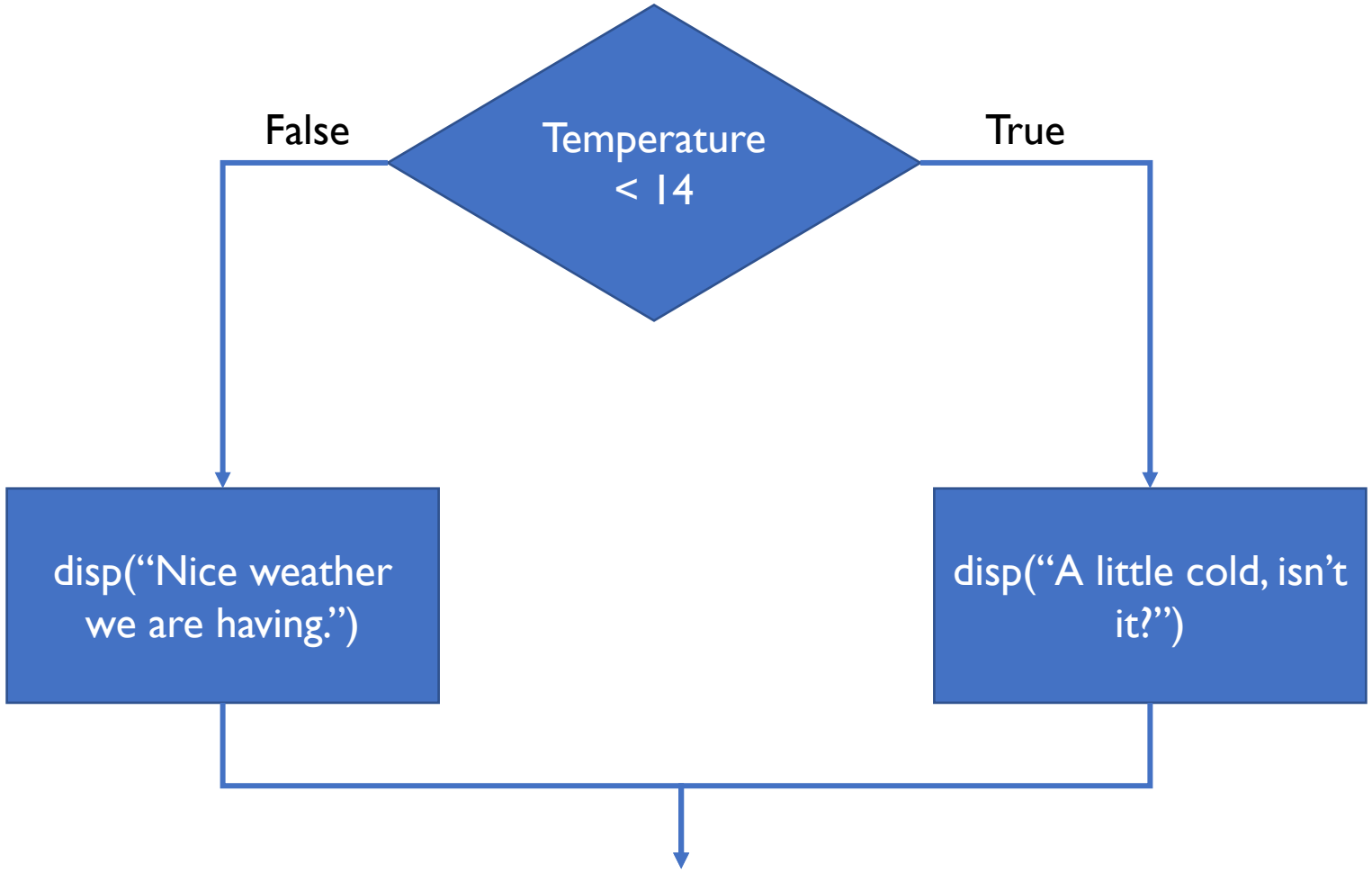
```
if course == "CSE101"
    mark = 100
end
```

# The if-else Statement

- Dual alternative decision structure: two possible paths of execution
  - One is taken if the condition is true, and the other if the condition is false
  - Syntax:

```
if condition
    statements
else
    statements
end
```

# If-else example



# Need to evaluate more conditions?

- Syntax:

```
if condition
    statements
elseif
    statements
elseif
    statements
else
    statements
end
```

# Example

```
x = 10;
minVal = 2;
maxVal = 6;

if (x >= minVal) && (x <= maxVal)
     disp('Value within specified range.')
elseif (x > maxVal)
     disp('Value exceeds maximum value.')
else
     disp('Value is below minimum value.')
end
```

# Practice 1

Write Matlab code to:

- Determine if someone qualifies for a loan, they must meet two conditions:
  - Must earn at least $50,000/year
  - Must have been employed for at least two years

- Check first condition, and if it is true, check second condition

# Building more complex conditions using logical operators

Recall

- Logical operators: operators that can be used to create complex Boolean expressions
  - and operator and or operator: binary operators, connect two Boolean expressions into a compound Boolean expression
  - not operator: unary operator, reverses the truth of its Boolean operand

## Aside: Truth tables

AND &

| a | b | a & b |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 0 |
| 1 | 0 | 0 |
| 1 | 1 | 1 |

OR |

| a | b | a \| b |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 1 |

Exclusive OR

| a | b | xor(a, b) |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 0 |

NOT ~

| a | ~a |
|---|---|
| 0 | 1 |
| 0 | 1 |
| 1 | 0 |
| 1 | 0 |

# Practice 2

- Modify the code from practice 1 to now use the logical operator AND.

# Next…

## Loops